

Biochip Course

Experimental Lab

Instructions

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a.y. 2023/2024

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Tasks (mandatory):

- Design electrodes and PCB (simulation of impedance)
- Perform measurements in liquid and data analysis

Tasks (optional):

- Circuit simulations (e.g. LTSpice)
- Design of fluidics

Expected learning outcomes:

- Develop hands-on experience
- Develop autonomous project organization skills
- Experience teamwork
- Develop presentation skills

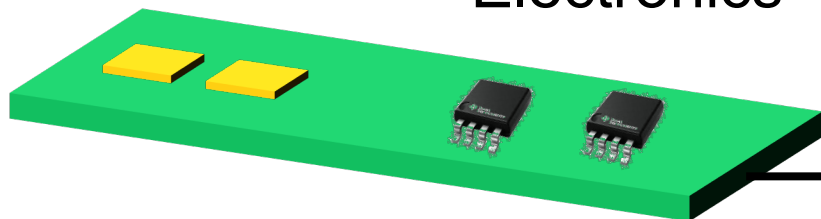


Experimental Lab Activity

Goal: design circuit to measure properties of liquid/solid interface

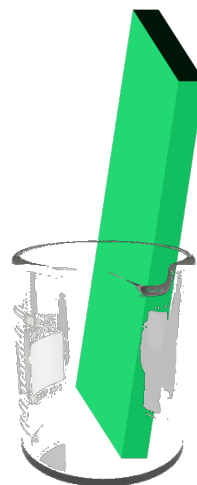
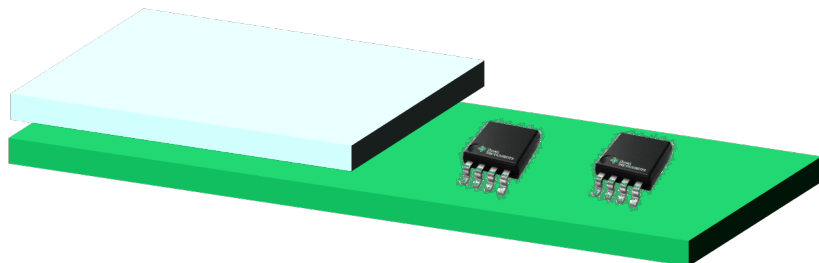
Electrodes

Front-End
Electronics



PCB

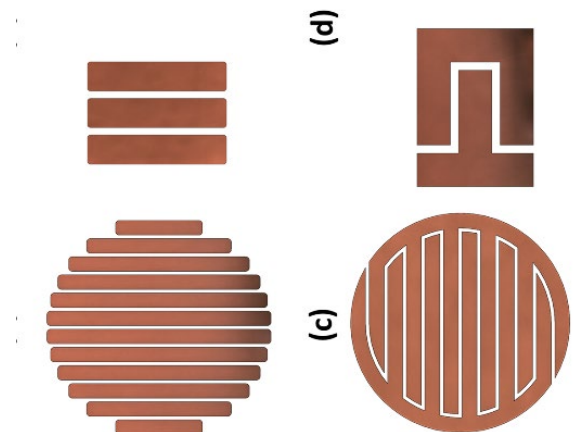
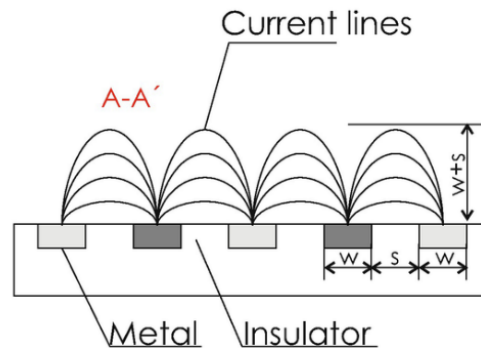
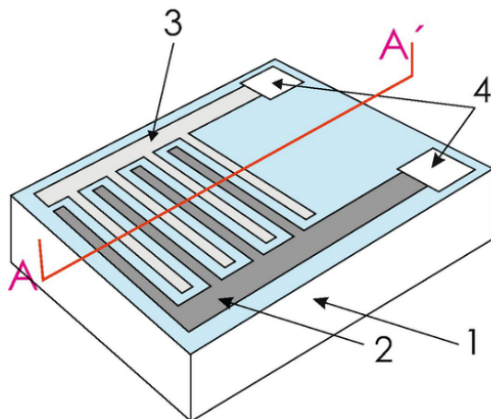
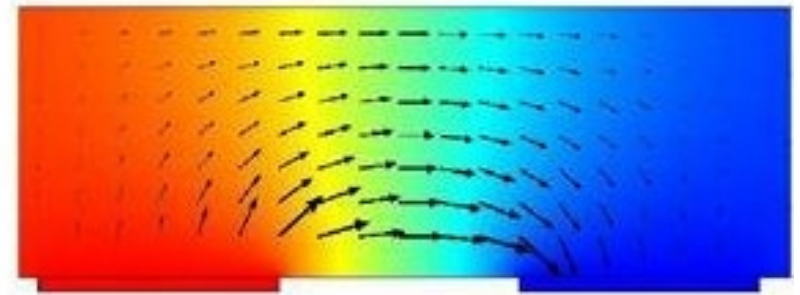
Microfluidics





1. Coplanar Electrodes

Electrodes will be gold-plated and coplanar: you can choose the size (area, distance) and shape





2. Estimation of Coplanar Impedance

Use conformal mapping to estimate R_{SOL} :



Available online at www.sciencedirect.com

SCIENCE @ DIRECT®

Sensors and Actuators A 112 (2004) 291–301



Analytical evaluation of the interdigital electrodes capacitance for a multi-layered structure

Rui Igreja*, C.J. Dias

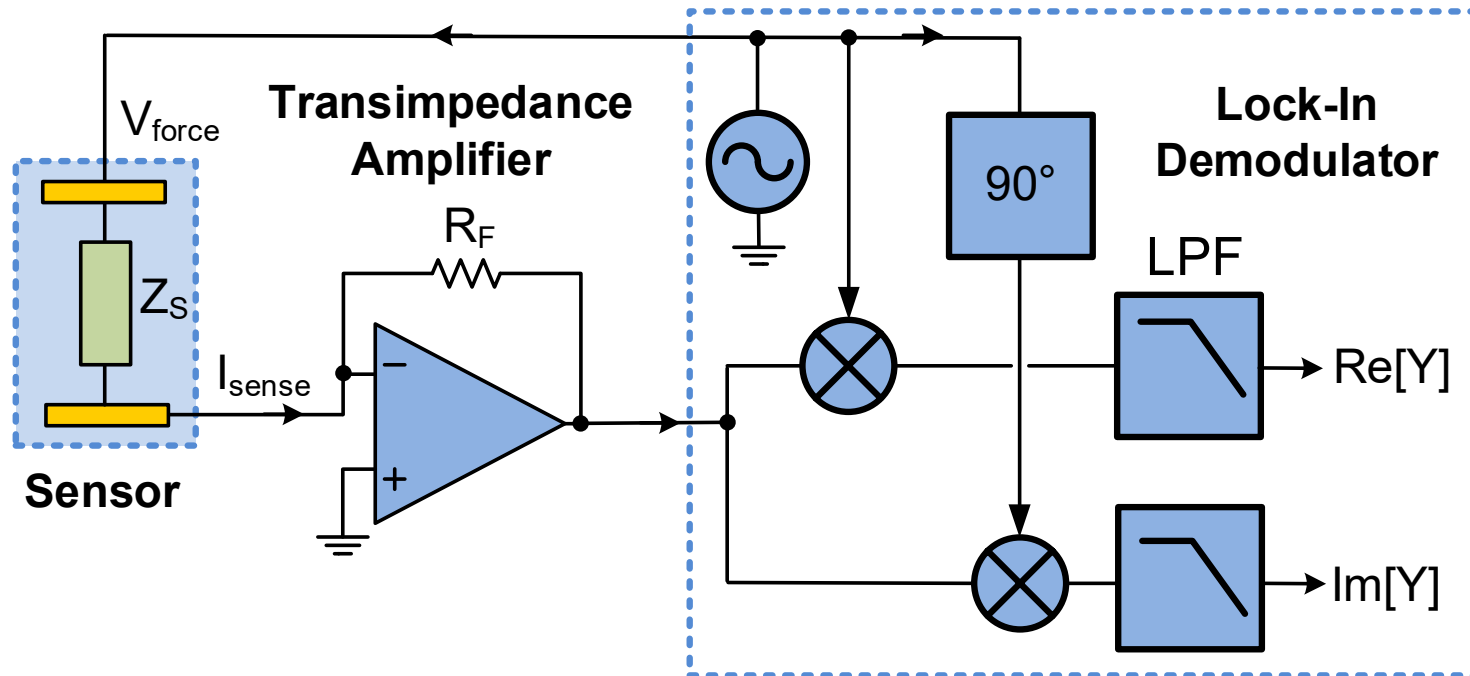
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Campus da Caparica, 2829-516 Caparica, Portugal*

Received 14 May 2003; received in revised form 14 January 2004; accepted 30 January 2004

- Starting point: tap water resistivity: in Milano $700\mu\text{S}/\text{cm}$
- Add ions (salt)
- Test also de-ionized water



3. Front-End Transimpedance Amplifier



Constraints:

- Dual voltage supply: $\pm 5\text{V}$
- Instrument (Moku:Go) input: $\pm 5\text{V}$
- OpAmp AD8022 (SOIC-8 SMT package)
- Spectroscopy f from 1kHz to 1MHz (10MHz), R_{SOL} visible



USB Instrument: Moku:Go Liquid Instruments

Analog Inputs/Outputs
Two 12 bit, 125 MSa/s

Input Bandwidth
30 MHz

Digital I/O
16 channel @ 62.5 MSa/s

Output Bandwidth
20 MHz

Programmable Power Supplies
2 or 4 channel option

11 Powerful Instruments

- Arbitrary Waveform Generator
- Data Logger
- Digital Filter Box
- Frequency Response Analyzer
- FIR Filter Builder
- Logic Analyzer / Pattern Generator
- Oscilloscope / Voltmeter
- PID Controller
- Spectrum Analyzer
- Waveform Generator
- Lock-in Amplifier* (add-on option)

Programmable Power Supplies

2 channel option

- +5 V to -5 V @ 150 mA
- 0 to 16 V @ 150 mA

4 channel option

- +5 V to -5 V @ 150 mA
- 0 to 16 V @ 150 mA
- Dual 0.6 to 5 V @ 1 A

Specifications

Analog Inputs

- Two 12 bit, 125 MSa/s input channels
- 30 MHz analog bandwidth
- AC or DC coupling with 1 M Ω impedance
- Input range up to ± 25 V

Analog Outputs

- Two 12 bit, 125 MSa/s output channels
- 20 MHz analog bandwidth
- ± 5 V maximum output range

Digital I/O

- 16-channel DIO at 62.5 MSa/s
- Support 3.3 V (5 V tolerant) logic level

Programming Environment

- API support for Python, MATLAB, and LabVIEW
- Windows or MacOS

Models

M0, \$599

- 2 analog inputs, 2 analog outputs and 16 DIO
- USB-C, Wi-Fi, software, and APIs

M1, \$699

- All features from M0
- Two-channel programmable power supply

M2, \$799

- All features from M0
- Ethernet
- Four-channel programmable power supply

Options & Accessories

- *Lock-in Amplifier, \$200
- All models include relevant accessories: 2 oscilloscope probes, DIO cabling, power adapter, USB-C and Ethernet cable, and power supply cables
- 6 standard colors, or custom color upgrade





2. Dual OpAmp



**ANALOG
DEVICES**

**Dual High Speed,
Low Noise Op Amp**

AD8022

FEATURES

Low power amplifiers provide low noise and low distortion,
ideal for xDSL modem receiver

Wide supply range: +5 V, ± 2.5 V to ± 12 V voltage supply

Low power consumption: 4.0 mA/Amp

Voltage feedback

Ease of Use

Lower total noise (insignificant input current noise
contribution compared to current feedback amps)

Low noise and distortion

2.5 nV/ $\sqrt{\text{Hz}}$ voltage noise @ 100 kHz

1.2 pA/ $\sqrt{\text{Hz}}$ current noise

MTPR < -66 dBc (G = +7)

SFDR 110 dB @ 200 kHz

High speed

130 MHz bandwidth (-3 dB), G = +1

Settling time to 0.1%, 68 ns

50 V/ μs slew rate

High output swing: ± 10.1 V on ± 12 V supply

Low offset voltage, 1.5 mV typical

FUNCTIONAL BLOCK DIAGRAM

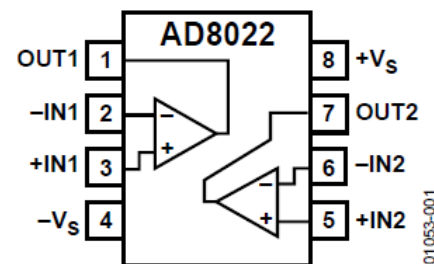


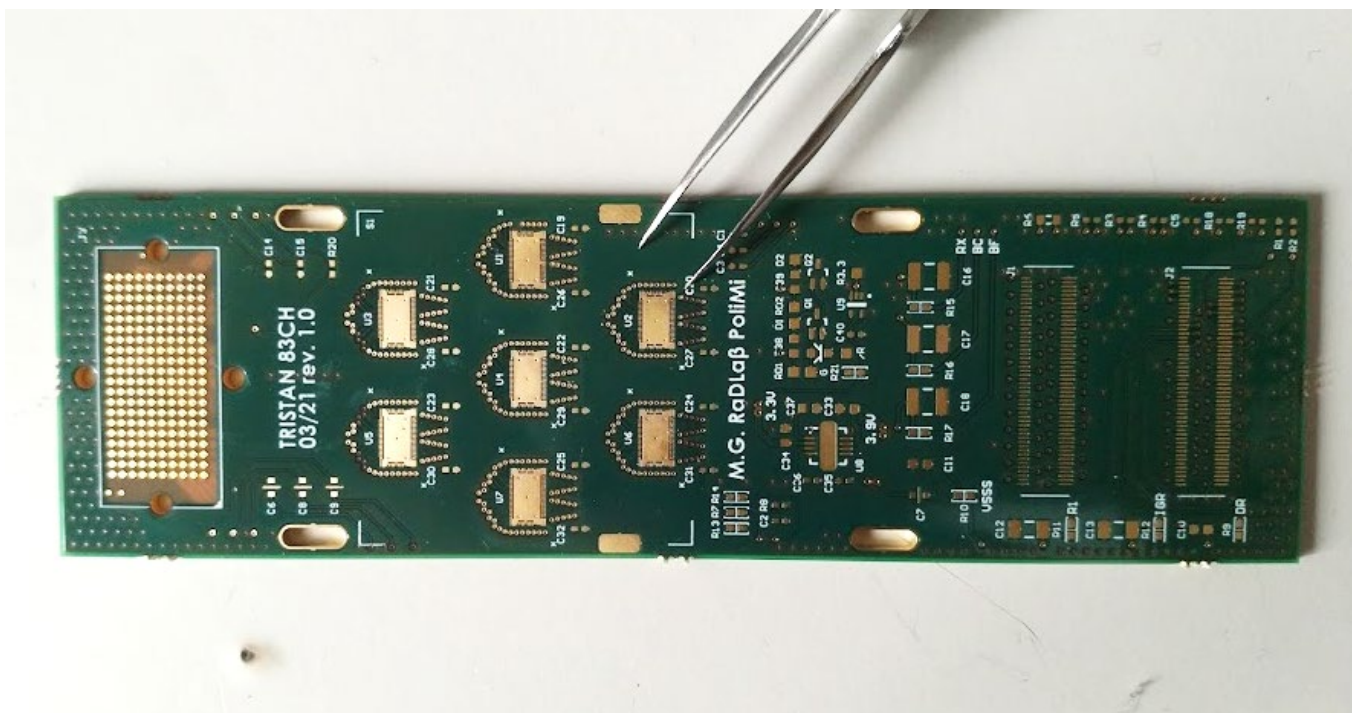
Figure 1.

Two opamps in one
package

Add decoupling
capacitors at V_s pins



2-Layer PCB



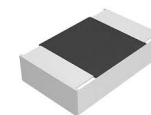
Gold
surface
finish



Board Size



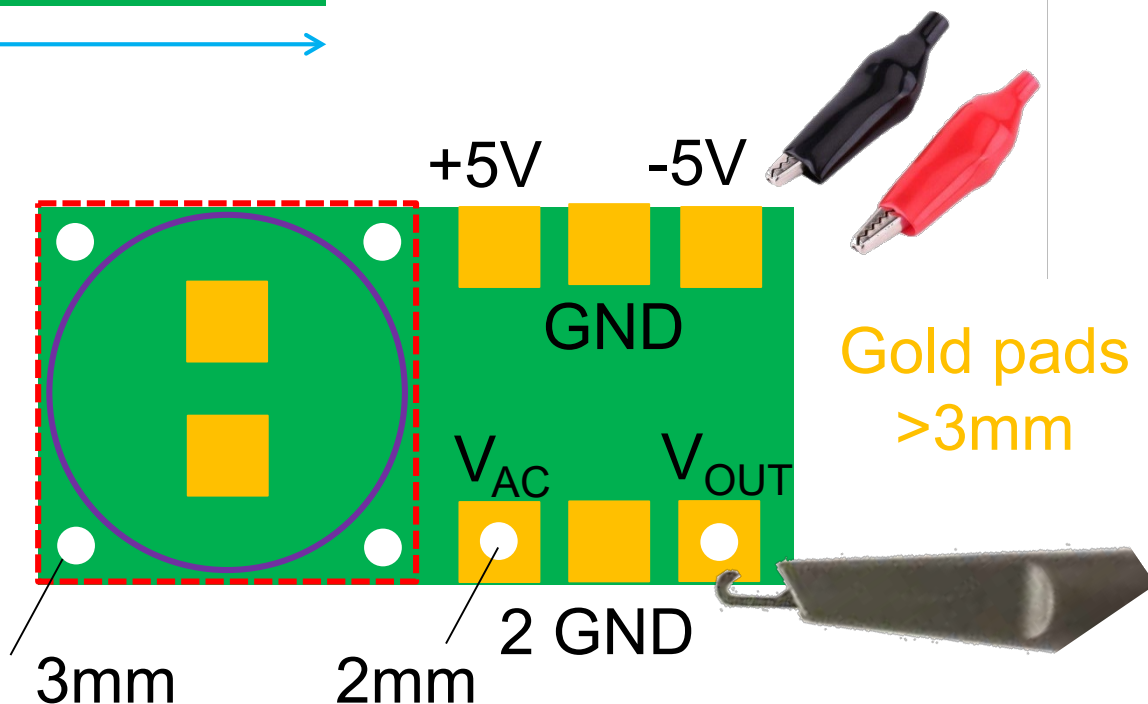
0805 SMD package
for passive
components



Pads for connections

Region to be coupled
with fluidics

O-ring 26mm ID,
28mm OD





PCB Design with KiCAD

Free software <https://www.kicad.org/download/>

The screenshot shows the KiCad website's download page. At the top is the KiCad logo and a navigation menu with links: BLOG, DISCOVER, COMMUNITY, HELP, CONTRIBUTE, SPONSORS, LIBRARIES, DOWNLOAD, and ABOUT. Below the menu is a banner image featuring a colorful PCB design. A breadcrumb trail reads 'Home / Download'. The main heading is 'Download', followed by the instruction 'Select your operating system'. Three large buttons are provided for selection: 'Windows' with the Windows logo, 'macOS' with the Apple logo, and 'Linux/Other' with the Tux penguin logo.

Tutorial:

<https://www.youtube.com/watch?v=aVUqaB0IMh4>



Project Already Created

Ki lab_2023 — KiCad 7.0

File Modifica Visualizza Strumenti Preferenze Aiuto



File progetto

- lab_2023.kicad_pro
 - lab_2023-backups
 - lab_2023_lib.pretty
 - lab_2023.kicad_pcb
 - lab_2023.kicad_sch



Editor degli schemi
Modifica lo schema elettrico del progetto



Editor simboli
Modifica le librerie di simboli globali e/o del progetto



Editor C.S.
Modifica il C.S. del progetto



Editor impronte
Modifica le librerie di impronte globali e/o del progetto



Visualizzatore Gerber



Project Already Created

lab_2023 — Editor C.S.

File Modifica Visualizza Inserisci Sbroglia Ispeziona Strumenti Preferenze Aiuto

Pista: usa larghezza netclass Via: usa dimensioni netclass Griglia: 2.5400 mm (0.1000 in) Zoom 8.00

F.Cu (PgUp)

Aspetto

Strati Oggetti

- F.Cu
- B.Cu
- F.Silkscreen
- B.Silkscreen
- F.Mask
- B.Mask
- Edge.Cuts
- Margin
- F.Courtyard
- B.Courtyard
- F.Fab
- B.Fab

Opzioni di visualizzazione

Preimpostazioni (Ctrl+Tab):

Viste (Shift+Tab):

Filtro selezione

- ☒ Tutti gli elementi
- ☒ Impronte
- ☒ Piste
- ☒ Piazzole
- ☒ Zone
- ☒ Dimensioni
- ☒ Elem
- ☒ Testc
- ☒ Via
- ☒ Grafi
- ☒ Aree
- ☒ Altri

Piazzole 19 Via 0 Segmenti di pista 0 Piste 9 Non sbrogliato 10

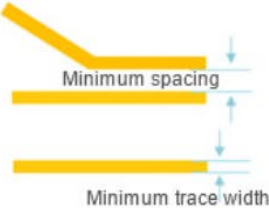
Z 7.06 X 182.8800 Y 83.8200 dx 182.8800 dy 83.8200 dist 201.1738 grid X 2.5400 Y 2.5400 mm



PCB Design Rules:

Minimum trace width and spacing



	Min. Trace width	Min. Spacing	Patterns
1-2 Layers	5mil (0.127mm)	5mil (0.127mm)	
4-6 Layers	3.5mil (0.09mm)	3.5mil (0.09mm)	
2oz Copper weight	8mil (0.2mm)	8mil (0.2mm)	

<https://jlcpcb.com/capabilities/Capabilities>



Fluidic Design (Optional)



Form 3 by FormLabs

25 μm resolution (x, y and z)

(take some margins)

Transparent resin

**Suggested software:
Fusion360 by Autodesk**



submit .stl file



Mandatory:

- Choose electrodes size and geometry
- Estimate impedance (by conformal mapping)
- Design TIA schematic (size R_F) and PCB layout
- Present and discuss your design
- Perform measurements in lab (changing ions concentration)

Optional:

- Spice simulations of the circuit (noise... [LTSpice](#))
- Propose **variants** (geometry, fluidics, 2 opamps...)
- Design of fluidics to be coupled to PCB
- Fitting of data for analysis



Deadlines

marzo 2024	aprile 2024	maggio 2024
1 ven	1 lun Lunedì dell'Angelo	1 mer
2 sab	2 mar	2 gio
3 dom	3 mer	3 ven
4 lun	4 gio	4 sab
5 mar	5 ven	5 dom
6 mer MI	6 sab	6 lun
7 gio POLI	7 dom	7 mar
8 ven	8 lun	8 mer
9 sab	9 mar MI	9 gio
10 dom	10 mer POLI	10 ven
11 lun	11 gio	11 sab
12 mar	12 ven	12 dom
13 mer	13 sab	13 lun
14 gio	14 dom	14 mar
15 ven	15 lun	15 mer
16 sab	16 mar	16 gio
17 dom	17 mer	17 ven
18 lun	18 gio	18 sab
19 mar	19 ven	19 dom
20 mer	20 sab	20 lun Lab 1
21 gio	21 dom	21 mar
22 ven	22 lun Discussion	22 mer Lab 2
23 sab	23 mar Fest. Ebr	23 gio
24 dom	24 mer Upload	24 ven
25 lun	25 gio	25 sab
26 mar	26 ven	26 dom
27 mer Start	27 sab	27 lun
28 gio	28 dom	28 mar
29 ven	29 lun Fest. Ebr	29 mer
30 sab	30 mar Fest. Ebr	30 gio
31 dom Pasqua		31 ven

Deadline:

- April 2nd: team formation
- April 22nd and 24th class discussion
5-min presentation per team + 5-min discussion
Max. 5 slides
- April 24th: projects submission
- May 20 & 22: lab activity



What to Address in the Presentation

Mandatory:

- Electrodes geometry
- Expected value of ionic resistance and spectrum
- Choice of R_F
- Ideas for test protocol to measure R_{ion}

Optional:

- Spice simulations of TIA
- Noise analysis
- Fluidics
- Other ideas



9 Teams & Submission

Register Team:

[BIOCHIP Lab Teams 2024.xlsx](#)

Teams:

1. Check teams members
2. Choose a name for each team
3. Create a group with teams members and start interacting

Submission Checklist:

1. **KiCAD file with layout** (no gerber)
2. **Value of R_F** and additional components if required
3. Optional: stl file for fluidics (or Fusion project)



Building 23, via Golgi, 42 (in front of admin offices)

